DOD-STD-1690(SH)
30 May 1979

METRIC

MILITARY STANDARD

MARITIME METRIC PRACTICE GUIDE

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DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

WASHINGTON, DC 20362

Maritime Metric Practice Guide

DOD-STD-1690(SH)

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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FOREWORD

The Appendix to this standard was developed by the Maritime Administration of the U.S. Department of Commerce, with the active participation of the Metric Practice Guide Task Group of the Society of Naval Architects and Marine Engineers and of metric representatives of the National Bureau of Standards and the Naval Sea Systems Command.

Rather than reformatting the end product of the above-cited effort, it has been attached intact as an Appendix, with errata sheets attached.

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1. SCOPE

1.1 Scope. This standard discusses SI units and those non-SI units accepted for limited use or time with SI, with particular application to the marine industry.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. Not applicable.

2.2 Other publications. See page 13 of appendix.

3. DEFINITIONS

3.1 SI. The International System of Units (from the French "Le Systeme International d'Unites"), as modified for use in the United States by the Secretary of Commerce (see also page 3 of appendix).

4. GENERAL REQUIREMENTS

4.1 General. Uniform metric practice for naval architecture, marine engineering, and general ship usage shall be in accordance with the appendix and errata sheets attached.

Preparing activity:
Navy - SH
(Project MISC-ND27)

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APPENDIX

MARITIME METRIC PRACTICE GUIDE

FOREWORD

The Metric Conversion Act of 1975 (P.L. 94-168) was signed into law by the President on December 23, 1975. The Act declared a national policy of coordinating the increasing use of the metric systems and established the United States Metric Board to coordinate voluntary conversion to a new system of measurement, the International System of Units, SI.[1]

This *Guide* will help obtain uniform metric practice in the marine industry by providing a technical reference for the International System of Units (SI). The *Guide* is not intended to cover all aspects of SI usage, but to serve as a ready reference especially tailored to the operating needs of the industry. For further information on SI usage and conversion factors for units not found here, the reader is referred to ANSI/ASTM E 380-76 "Standard for Metric Practice" (approved for use by agencies of the Department of Defense)[2] upon which this *Guide* is based.[3] Hardware and other standards in SI metric are currently being developed. Information on their status may be obtained from the appropriate standards-writing organization.

The *Guide* contains discussions of the base, supplementary, and derived SI units. Also discussed are non-SI units which have been accepted and recognized by the CGPM as appropriate for limited use or time. Basic rules for style and usage of SI are set forth, as well as methods for conversion from inch-pound units to SI units. A table of quantities used by the marine industry is included, with present units, the preferred SI units, and conversion factors given.

1.0 SI UNITS

SI includes three classes of units:

base units
supplementary units
derived units

---

[1] SI was adopted in 1960 by the General Conference on Weights and Measures (CGPM) of which the United States is a member.

[2] Reference A.

[3] See also Reference B, published pursuant to the authority given the Secretary of Commerce under the Act to intercept and modify the SI for use within the United States.

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1.1 Base Units

The International System of Units is based on seven units, listed in Table 1, which by convention are regarded as dimensionally independent.

Table I—SI BASE UNITS

Quantity	Base SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

1.2 Supplementary Units

The units listed in Table II are called supplementary units and may be regarded as base units or as derived units.

Table II—SI SUPPLEMENTARY UNITS

Quantity	Supplementary SI Unit	Symbol
Plane angle	radian	rad
Solid angle	steradian	sr

1.3 Derived Units

Derived units are formed by the algebraic combination of base units, supplementary units, and other derived units. The derived units with special names which have been approved for use are listed in Table III.

1.4 Multiples of SI Units

The prefixes and symbols shown in Table IV are used to form names and symbols of the decimal multiples and submultiples of the SI units.

1.5 Selection of Prefixes

1.5.1 A prefix should be selected so that the numerical value of the unit expressed will fall between 0.1 and 1000. An exception to this general rule arises in the preparation of tables of values of the same quantity and in discussion of such values within a given context, when it is better to use the same unit multiple. Also, for certain applications, one particular multiple will customarily be used; for example, use of the millimeter for linear dimensions in engineering drawings.

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1.5.2 Compound prefixes should not be used; for example, use GJ, not kMJ.

1.5.3 Prefixes should preferably not be used in the denominator of compound units. For example, use V/m not mV/mm. The exception to this occurs when the kilogram is one of the units: MJ/kg, not kJ/g.

1.5.4 Errors in calculation may be avoided by using powers of ten with the units rather than prefixes. For example, 6 m x 4 m x 15 mm may better be calculated from $6 \text{ m} \times 4 \text{ m} \times 15 \times 10^{-3} \text{ m} = 0.36 \text{ m}^3$

Table III—SI DERIVED UNITS WITH SPECIAL NAMES

Quantity	Name of derived SI unit	Symbol	Expressed in terms of Base or supplementary SI units or in terms of other derived SI units
Frequency	hertz	Hz	$1 \text{ Hz} = 1 \text{ s}^{-1}$
Force	newton	N	$1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2$
Pressure, stress	pascal	Pa	$1 \text{ Pa} = 1 \text{ N/m}^2$
Energy, work, quantity of heat	joule	J	$1 \text{ J} = 1 \text{ N}\cdot\text{m}$
Power, radiant flux	watt	W	$1 \text{ W} = 1 \text{ J/s}$
Electric charge, quantity of electricity	coulomb	C	$1 \text{ C} = 1 \text{ A}\cdot\text{s}$
Electric potential, potential difference, electromotive force	volt	V	$1 \text{ V} = 1 \text{ W/A}$
Capacitance	farad	F	$1 \text{ F} = 1 \text{ C/V}$
Electric resistance	ohm	Ω	$1 \Omega = 1 \text{ V/A}$
Conductance	siemens	S	$1 \text{ S} = 1 \text{ A/V}$
Magnetic flux	weber	Wb	$1 \text{ Wb} = 1 \text{ V}\cdot\text{s}$
Magnetic flux density	tesla	T	$1 \text{ T} = 1 \text{ Wb/m}^2$
Inductance	henry	H	$1 \text{ H} = 1 \text{ Wb/A}$
Luminous flux	lumen	lm	$1 \text{ lm} = 1 \text{ cd}\cdot\text{sr}$
Illuminance	lux	lx	$1 \text{ lx} = 1 \text{ lm/m}^2$
Celsius temperature interval	degree Celsius	°C	$1 \text{ °C} = 1 \text{ K}$
Activity (of a radionuclide)	becquerel	Bq	$1 \text{ Bq} = 1 \text{ s}^{-1}$
Absorbed dose	gray	Gy	$1 \text{ Gy} = 1 \text{ J/kg}$

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Table IV—SI PREFIXES

Prefix	Symbol	Factor
exa	E	10^{18} 1 000 000 000 000 000 000
peta	P	10^{15} 1 000 000 000 000 000
tera	T	10^{12} 1 000 000 000 000
giga	G	10^{9} 1 000 000 000
mega	M	10^{6} 1 000 000
kilo	k	10^{3} 1 000
hecto	h	10^{2} 100
deka	da	10^{1} 10
deci	d	10^{-1} 0.1
centi	c	10^{-2} 0.01
milli	m	10^{-3} 0.001
micro	μ	10^{-6} 0.000 001
nano	n	10^{-9} 0.000 000 001
pico	p	10^{-12} 0.000 000 000 001
femto	f	10^{-15} 0.000 000 000 000 001
atto	a	10^{-18} 0.000 000 000 000 000 001

2.0 ACCEPTABLE NON-SI UNITS

2.1 Units Accepted for Limited Purposes[4]

Certain units which are not a part of SI have been accepted for use with SI units. These units, listed below in Tables V and VI are the only units currently recognized as acceptable for use with SI.

Table V—OTHER UNITS ACCEPTED FOR USE WITH SI

Quantity	Unit	Symbol	Definition
time	minute	min	1 min = 60 s
	hour	h	1 h = 60 min = 3600 s
	day	d	1 d = 24 h = 86 400 s
plane angle	degree	°	1° = (π/180) rad
	minute	′	1′ = (1/60)° = (π/10 800) rad
	second	″	1″ = (1/60)′ = (π/648 000) rad
area	hectare	ha	1 ha = 1 hm^2 = 10^4 m^2
volume	liter	L	1 L = 1 dm^3 = 10^{-3} m^3
mass	metric ton	t	1 t = 10^3 kg

Time—The SI unit of time is the second. This unit is preferred and should be used when practical, particularly in technical calculations.

[4] Reference B.

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Plane Angle—The SI unit of plane angle is the radian. When the radian is not a convenient unit, the degree should be used with decimal submultiples. Minutes and seconds should be used only when required (as in navigation).

Area—The SI unit of area is the square meter. Large areas are properly expressed in square hectometers or square kilometers. The hectare (used as an alternative to the square hectometer), is restricted to the measurement of land or water areas.

Temperature—The SI unit of thermodynamic temperature is the kelvin, and this unit is properly used for expressing thermodynamic temperature and temperature intervals. The degree Celsius is also an SI unit and is properly used for expressing Celsius temperature and temperature intervals. Celsius temperature t equals the difference ($t=T-T_o$) between the two thermodynamic temperatures T and T_o where $T_o=273.15$ K.

Volume—The SI unit of volume is the cubic meter. The cubic meter, or one of its multiples or submultiples, is preferred for all applications. The special name liter has been approved by the CGPM for the cubic decimeter, but is restricted to the measurement of liquids and gases. No prefixes other than "milli-" or "micro-" should be used with liter.

Mass—The SI unit of mass is the kilogram. The kilogram, or one of the multiples or submultiples formed by attaching an SI prefix to gram, is preferred for all applications. For large masses (such as have been expressed in tons), the megagram is the appropriate unit. The term metric ton should be restricted to commercial and maritime usage, and no prefixes should be used with it. To avoid confusion, use of the term "tonne" to indicate metric ton is discouraged. (See also 3.0)

Table VI—OTHER UNITS ACCEPTED FOR LIMITED USE

Because their usage is already well established, these units may be used subject to further review.

nautical mile (nmi)	1.852 km
knot (kn)	nmi/hr.
bar	100 kPa
gal[1]	1 cm/s²

[1] unit of acceleration

3.0 MASS, FORCE AND WEIGHT

One of the major departures SI has taken from the older metric systems is the use of distinctly separate units for mass and force.

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In SI, the unit of force, the newton (N), instead of being related to gravity, is defined as being equal to the acceleration it imparts to a unit mass, the kilogram (kg). This should eliminate confusion caused by earlier use of the kilogram-force as the unit of force while the kilogram was used for the mass unit.

3.1 Mass

The SI unit for mass (not force) is the kilogram, used to specify the quantity of matter in a body. The mass of a body never varies and is independent of gravitational forces.

3.2 Force

The SI unit (derived) for force is the newton. One newton applied to a mass of one kilogram gives an acceleration of one meter per second squared ($N = kg \cdot m/s^2$). The newton is used in all combinations which include force, such as pressure and stress (N/m^2), energy ($N \cdot m$), and power ($N \cdot m/s = W$).

3.3 Weight

Weight is sometimes defined as the force which, when applied to a body, would give it an acceleration equal to the local acceleration of free fall. However, this technical use of the term is generally disregarded in commercial and everyday use, when reference to the weight of a body is used to indicate its mass. Because of the dual use of the term weight as a quantity, this term should be avoided in technical practice unless its meaning is made completely clear. When the term weight is used, it must be known whether mass or force is intended and the appropriate SI units utilized.

4.0 RULES FOR STYLE AND USAGE OF SI

4.1 Rules for Writing Unit Symbols

4.1.1 Particular care must be taken to use the correct case for symbols, units, and their multiples (e.g., K for kelvin, k for kilo, M for mega, m for milli). When using systems with limited character sets, as in Telex transmis- be used. For these purposes, refer to ISO 2955 (1974)[5] or ANSI X3.50-1976.

4.1.2 Unit symbols are always written in the singular form.

4.1.3 Unit symbols should be printed in roman (upright) type, regardless of the type style used in the surrounding text.

[5] Reference C.

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4.1.4 Unit symbols are not followed by a period except when used at the end of a sentence.

4.1.5 The numerical value associated with a symbol should be separated from that symbol by a space. For example 25.4 mm, not 25.4mm. The only exception to this rule is that no space is left between the numerical value and the symbols for degree, minute, and second of plane angle and degree Celsius.

4.1.6 Unit symbols should be used in preference to the unit names except when a number written out in words precedes the unit; for example "seven meters" not "seven m".

4.2 Rules for Writing Unit Names

4.2.1 The first letter of a unit name is not capitalized except at the beginning of a sentence or in capitalized material such as a title.

4.2.2 Plurals of unit names are formed in the ordinary manner, except for lux, hertz, and siemens, which remain the same.

4.2.3 No space or hyphen is used between a prefix and the unit name; for example, kilonewton.

4.2.4 The spelling "meter" and "liter" as used in this publication are preferred. However, "metre" and "litre" are also widely used.

4.3 Product, Quotient and Powers

4.3.1 To indicate the product of units when using their names, a space is left between the names (e.g. newton meter). When using symbols, a centered dot should be placed between the symbols (e.g. N•m).

4.3.2 To indicate the quotient of units when using their names use the word "per" (e.g. meters per second). When using unit symbols, a solidus (/) or negative exponent should be used (e.g. m/s or $m \cdot s^{-1}$). Do not use more than one solidus in the same expression unless parentheses are inserted to avoid ambiguity.

4.3.3 To indicate powers when using unit names, the words "square", "cubic", "squared", "cubed", etc. should be used (e.g. square meter, second squared). When using unit symbols, powers are indicated by the use of an exponent (e.g. m^3).

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4.4 Numbers

4.4.1 As at present, the period will be used for the decimal marker. For numbers of less than one, a zero should precede the decimal; for example 0.964.

4.4.2 Recommended international practice calls for the use of a space to separate large numbers into groups of three. Do not use a comma. The digits should be separated into groups of three counting from the decimal point toward the left and the right, and using a small space to separate the groups. In numbers of four digits, the space is usually not necessary except to provide uniformity in tables, columns, etc. For example, 16 390, 1.828 8 or 1.8288, 4 536 or 4536.

5.0 CONVERSION AND ROUNDING METHODS

5.1 Soft Conversion

Soft conversion is the process of changing the inch-pound measurement language to equivalent metric units within acceptable measurement tolerances without changing the physical configuration. In other words, it is the same item both before and after conversion.

5.2 Hard Conversion

Hard Conversion is the process of changing the inch-pound measurement language to non-equivalent metric units which necessitates physical configuration changes outside those permitted by established measurement tolerances. Although the term is in general use, it is technically incorrect when applied to specific items because no "conversion" takes place; rather, a new metric item (requiring a new part identification) is created to replace the customary item.

5.3 Conversion Factors

Table VII contains conversion factors for the units commonly used in the marine field.

5.4 Precision

When converting, care must be given to the precision desired. The number of significant digits retained in the answer should be such that accuracy is neither sacrificed nor exaggerated. Specified quantities should first be multiplied by the exact conversion factor and then rounded to the appropriate number of significant digits. For example, to convert 40 ft^3 to meters:

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40 x 0.028 316 85 = 1.132 674 which rounds to 1.133 m^3. Do not round either the conversion factor or the quantity before performing the multiplication, as accuracy would be reduced.

5.5 Rounding

The rounding required by conversion and computation in SI units should be done using the following rules for the three general cases:[6]

a) Where the digit immediately following the last digit to be retained is less than 5, the last digit retained should not be changed.

b) Where the digit immediately following the last digit to be retained is greater than 5 (or it is a 5 followed by at least one digit other than 0), the last digit should be increased by one.

c) Where the digit immediately following the last digit to be retained is exactly 5 (or it is a 5 followed by only zeros) the last digit retained is not changed if it is even, and if it is odd it should be increased by one. For example, 4.365 00 becomes 4.36 when rounded to three digits. 4.355 00 also becomes 4.36 when rounded to three digits.

The above rules provide an additional observation about significant digits. The least significant digit (the last digit retained in the rounding process) cannot be considered more precise than 0.5 of its place value.

6.0 ACCURACY OF CONVERSION

It is important, in any conversion problem, to determine the accuracy of the original value and the repeatability of the converted value. Where interchangeability of converted values is required, ANSI/ASTM E 380-76[7] should be consulted for exact procedures which will establish limits to the errors resulting from conversion calculations.

7.0 SPECIAL RATIOS

Two special ratios[8] used in the Maritime Industry are based on dimensionless numbers but are currently used in dimensional form.

[6] Adapted from Reference D.
[7] Reference A.
[8] Reference E

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These ratios are speed-length ratio, $V/\sqrt{L}$, and displacement-length ratio, $\triangle/(0.01L)^3$, where

V is speed in knots
L is length in feet, and
$\triangle$ is displacement in long tons

When using metric units, the speed-length ratio should be replaced by Froude number, $v/\sqrt{gL}$, in consistent units, where

v is velocity
g is gravitational acceleration, and
L is the ship's length

Similarly the displacement-length ratio should be replaced by the fatness ratio[9], $\nabla/(0.10L)^3$, in consistent units, where

∇ is the underwater volume of the hull.
L is the ship's length

Speed-length ratio and displacement-length ratio are in common use because they have convenient values in their dimensional form, e.g., $V/\sqrt{L} = 1$ is significant from a wave resistance viewpoint. Using metric units would cause the numerical values of speed-length raito to be increased by a factor of 1.81 and displacement-length ratio by 35.9.

Froude number is a factor of 0.298 times speed-length ratio. Since using metric units in speed-length ratio or using Froude number will change the values used, it is logical to use a proper dimensionless ratio, i.e., Froude number. Thus, Froude number should be used in lieu of speed-length ratio when designing in the metric system.

Two dimensionless ratios. [10] volumetric coefficient[11] ($C_v = \nabla/L^3$) and fatness ratio, are in common use and are equivalent to the displacement-length ratio. They are 0.000035 times and 0.035 times the displacement-length ratio, respectively (assuming the displacement is in salt water). Thus, either use of a metric version of the displacement-length ratio or use of C_v, or fatness ratio involves a significant shift in the numerical values which will occur in practice. Further, C_v results in much smaller numbers than fatness ratio. Since fatness ratio is already in current use and it results in more reasonable values than C_v, standardization on it to the exclusion of displacement-length ratio is recommended when designing in the metric system.

[9] Reference F
[10] Reference G
[11] Reference H

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REFERENCES

A. ANSI/ASTM E 380-76 "Standard for Metric Practice" (American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

B. FR Doc. 77-31094 "The Metric System of Measurement—Interpretation and Modification of the International System of Units for the United States". October 26, 1977.

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TABLE VII

UNITS

APPLICATIONS

CONVERSION FACTORS

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CONVERSION TABLE

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present(2)
1-1.1	Plane Angle				
	Machine design	radian	radian (rad)	1.0*	1.0*
	Angle of entrance of hull, engine timing, navigation	degree	degree (°)*	1.0*	1.0*
		minute	degree (°)	1 min(3) = 0.016 667 deg.	1 degree = 60 min.
		second	degree (°)	1 sec(3) = 0.000 2778 deg.	1 degree = 3600 sec.
1-3.1	Length				
	Distances	nautical mile(4)	nautical mile	1.0*	1.0*
		statute mile	kilometer (km)	1 statute mile = 1.6093 km.	1 km = 0.62 statute mile
	Depth of water, length of cables, wires and ropes	fathom	meter (m)	1 fathom = 1.8288 m*	1 m = 0.55 fathom
	Deck heights, draft, main hull dimensions, stability data, location of centers of gravity, buoyancy and flotations, KM, KG, GM, and GZ	foot & inch	meter (m)	1 foot = 0.3048 m*	1 m = 3.28 foot
	Draft marks on the hull	foot	decimeter (dm)	1 foot = 3.048 dm*	1 dm = 0.33 foot
	Deck camber, frame scantlings and plate thickness, insulation thickness, machinery dimensions, pipe sizes	foot and/or inch	millimeter (mm)	1 foot = 304.8 mm* 1 inch = 25.4 mm*	1 mm = 0.0033 foot 1 mm = 0.039 inch

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CONVERSION TABLE—Continued

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present(2)
	Rope & cable diameters	inch	millimeter (mm)	1 inch = 25.4 mm*	1 mm = 0.039 inch
	Paint film thickness	10^{-3} inch (mil)	micrometer (μm)	1 mil = 25.4 μm*	1 μm = 0.039 mil
1-4.1	Area				
	Deck areas, heating surface, panelling areas, propeller blade area, waterplane area, wetted hull surface, free surface	square yard	square meter (m^2)	1 yd^2 = 0.836 127 m^2	1 m^2 = 1.196 yd^2
		square foot	square meter (m^2)	1 ft^2 = 0.092 903 m^2	1 m^2 = 10.76 ft^2
	Sectional areas of stressed parts	square inch	square millimeter (mm^2)	1 in^2 = 645.16 mm^2*	1 mm^2 = 0.0016 in^2
1-4.1a	First Moment of Area				
	Structure design, midship section	$inch^3$ foot	meter centimeter^2 ($m \cdot cm^2$)	1 in^2 ft = 1.966 488 $m \cdot cm^2$	1 $m \cdot cm^2$ = 0.51 in^2 ft
1-4.1b	Section Modulus				
	Frames and associated plating	$inch^3$	millimeter cubed (mm^3)	1 in^3 = 16 387 .06 mm^3	1 mm^3 = 6.1 x 10^{-5} in^3
		$inch^2$ foot	meter centimeter^2 ($m \cdot cm^2$)	1 in^2 ft = 1.966 488 $m \cdot cm^2$	1 $m \cdot cm^2$ = 0.51 in^2 ft
1-4.1c	Second Moment of Area				
	Structure design, midship section	$inch^2$ $foot^3$	meter^{2}centimeter^3 ($m^2 \cdot cm^3$)	1 in^2 ft^3 = 0.599 373 $m^2 \cdot cm^3$	1 $m^2 \cdot cm^3$ = 1.668 in^2 ft^3
	Frames and associated plating	$inch^4$	millimeter^4 (mm^4)	1 in^4 = 416 231.4 mm^4	1 mm^4 = 2.4 x 10^{-6} in^4

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1-5.1 Volume

Freight volume measurement(5)	ton of 40 ft³	cubic meter (m³)	1 freight ton = 1.132 67 m³	1 m³ = 0.883 freight ton
Gross and net tonnage (measurement ton)	ton of 100 ft³	cubic meter (m³)	100 ft³ = 2.831 685 m³	1 m³ = 35.31 measurement ton
Volume of displacement, volume of holds, large air receivers	cubic foot	cubic meter (m³)	1 ft³ = 0.028 317 m³	1 m³ = 35.31 ft³
Volume of large fuel tanks	barrel (42 gallon)	cubic meter (m³)	1 bbl = 0.158 987 m³	1 m³ = 6.29 bbl
Volume of small tanks and small air receivers	U. S. gallon (liquid)	liter (L)	1 gal = 3.785 412 L	1 L = 0.26 gal
	U. S. gallon (liquid)	cubic meter (m³)	1 gal = 0.003 785 m³	1 m³ = 264.2 gal

1-5.1a Specific Volume

Stowage factors for solid and liquid cargo, fuel oil and water	foot³/ton	cubic meter per metric ton (m³/t)	1 ft³/ton = 0.027 870 m³/t	1 m³/t = 35.88 ft³/ton
Gases and vapors	foot³/pound	liter per kilogram (L/kg)	1 ft³/lb = 62.427 97 L/kg	1 L/kg = 0.016 ft³/lb

1-5.1b Volume Flow

Capacity of air compressors, fans, large pumps	feet³/minute	cubic meters per hour (m³/h)	1 ft³/min = 1.699 m³/h	1 m³/h = 0.588 ft³/min
Capacity of most pumps	gallon/minute	liter per second (L/s)	1 gal/min = 0.063 090 L/s	1 L/s = 15.85 gal/min
			1 gal/min = 0.000 063 m³/s	1 m³/s = 15 873.0 gal/min
Barrel per day	bbl/day	cu. meter/day (m³/d)	1 bbl/d = 0.158 987 m³/d	1 m³/d = 6.29 bbl/d
Barrel per nautical mile	bbl/nmi	cu. meter/nmi (m³/nmi)	1 bbl/mi = 0.158 .987 m³/nmi	1 m³/nmi = 6.29 bbl/mi

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CONVERSION TABLE—Continued

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present(2)
1-6.1	Time				
	As at present	day (d)	day (d)	1.0*	1.0*
		hour (h)	hour (h)	1.0*	1.0*
		minute (m)	minute (min)	1.0*	1.0*
		second (s)	second (s)	1.0*	1.0*
1-8.1	Angular velocity				
	Machine design	radian/second	radian per second (rad/s)	1.0*	1.0*
1-8.1a	Angular acceleration				
	Machine design	radian/second^2	radian per second squared (rad/s^2)	1.0*	1.0*
1-10.1	Velocity				
	Ship's speed	knot(4)	knot	1.0*	1.0*
	Fluid velocity, linear velocity of machinery parts, speed of cables, ropes, wires	foot/second	meter per second (m/s)	1 ft/s = 0.3048 m/s*	1 m/s = 3.28 ft/s
1-11.1	Acceleration				
	Machine design Vibration	foot/second^2	meter per second squared (m/s^2)	1 ft/s^2 = 0.3048 m/s^2*	1 m/s^2 = 3.28 ft/s^2
2-3.1	Frequency				
	Electronics	cycle/sec	hertz (Hz)	1.0*	1.0*
2-3.2	Rotational Speed				
	Rotational speed of machinery	revolution per minute (r/min) (formerly rpm)	revolution per minute (r/min)	1.0*	1.0*

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3-1.1 Mass				
Deadweight, displacement, lightweight ballast, fresh water and fuel oil capacity	ton (long)([a])	metric ton (t)	1 ton = 1.016 047 t	1 t = 0.98 ton
Lifting capacity, test load and SWL (safe working load, rated) of cranes, booms, and winches	ton	metric ton (t)	1 ton = 1.016 047 t	1 t = 0.98 ton
	pound	kilogram (kg)	1 lb = 0.453 592 kg	1 kg = 2.20 lb
general stores, pieces of equipment	pound (lb)	kilogram (kg)	1 lb = 0.453 592 kg	1 kg = 2.20 lb
	hundred weight (cwt)	kilogram (kg)	1 cwt = 50.802 35 kg	1 kg = 0.02 cwt
3-1.1a Moment of Mass	pound foot	kilogram meter (kg•m)	1 lb-ft. = 0.138 255 kg•m	1 kg•m = 7.23 lb•ft
	ton foot	metric ton meter (t•m)	1 ton•ft. = 0.309 691 t•m	1 t•m 3.23 ton•ft
Trimming moments	ton foot	metric ton meter (t•m)	1 ton•ft. = 0.309 691 t•m	1 t•m = 3.23 ton•ft
3-1.1b Mass Flow				
Liquid flow rates	ton/hour	metric ton per hour (t/h)	1 ton/hour = 1.016 047 t/h	1 t/h = 0.98 ton/hour
Gas and steam flow rates	pound/hour	kilogram per hour (kg/h)	1 lb/h = 0.453 591 kg/h	1 kg/h = 2.20 lb/h
3-1.1c Displacement				
Displacement per unit of immersion	tons per inch	metric ton per centimeter (t/cm)	1 ton/in = 0.400 019 t/cm	1 t/cm = 2.5 ton/in
3-1.1d Moment to change trim by a unit amount	ton foot per inch	metric ton meter per centimeter (t•m/cm)	1 ton•ft/in = 0.121 926 t•m/cm	1 t•m/cm = 8.20 ton•ft/in

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CONVERSION TABLE—Continued

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present (2)
3-1.1e	Moment to change list by a unit amount	ton foot per degree	ton meter per degree (t•m/degree)	1 ton•ft/deg = 0.309 691 t•m/deg	1 t•m/deg = 3.23 ton•ft/deg
3-2.1	Density				
	Density of solids	pound/inch³	kilogram per cubic centimeter (kg/cm³)	1 lb/in³ = 0.027 680 kg/cm³	1 kg/cm³ = 36.13 lb/in³
	Density of liquids	pound/gallon	kilogram per liter (kg/L)	1 lb/gal = 0.119 826 kg/L	1 kg/L = 8.34 lb/gal
	Density of gases	pound/foot³	kilogram per cubic meter (kg/m³)	1 lb/ft³ = 16.018 46 kg/m³	1 kg/m³ = 0.062 lb/ft³
3-2.1a	Concentration				
	Salinity Boiler feed water testing, potable water testing	grains/gallon	grams per liter (g/L)	1 grain/gal = 0.17 118 g/L	1 g/L = 58.418 grain/gal
3-5.1	Momentum Machine design	pound foot/second	kilogram meter per second (kg•m/s)	1 lb•ft/s = 0.138 255 kg•m/s	1 kg•m/s = 7.23 lb•ft/s
3-6.1	Moment of Momentum (Angular Momentum) Machine design	pound foot²/second	kilogram meter squared per second (kg•m²/s)	1 lb•ft²/s = 0.042 140 kg•m²/s	1 kg•m²/s = 23.73 lb ft²/s

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3-7.1 Moment of Inertia				
Machine design	pound foot² ($kg \cdot m^2$)	kilogram meter² ($kg \cdot m^2$)	1 lb•ft² = 0.042 140 kg•m²	1 kg•m² = 23.73 lb•ft²
	pound inch² ($kg \cdot m^2$)	kilogram meter² ($kg \cdot m^2$)	1 lb•in² = 0.292 637 x 10^{-3} kg•m²	1 kg•m² = 3 417.2 lb•in²
3-8.1 Force				
Machine and structure design	ton-force	kilonewton (kN)	1 tonf = 9.064 017 kN	1 kN = 0.1 tonf
	pound-force	newton (N)	1 lbf = 4.448 222 N	1 N = 0.225 lbf
3-10.1 Moment of Force (Torque)				
Machine and structure design, bending moments	foot ton-force	kilonewton meter (kN•m)	1 tonf•ft = 3.037 032 kN•m	1 kN•m = 0.33 tonf•ft
	foot pound-force	newton meter (N•m)	1 lbf•ft = 1.355 818 N•m	1 N•m = 0.74 lbf•ft
	inch pound-force	newton meter (N•m)	1 lbf•in = 0.112 985 N•m	1 N•m = 8.85 lbf•in
	foot poundal	newton meter (N•m)	1 pdl•ft = 0.042 140 N•m	1 N•m = 23.73 pdl•ft
Torque(7)				
3-11.1 Pressure				
Normal fluid pressure, e.g., compressed air, oil, steam	pound-force/inch²(8)	kilopascal (kPa)	1 lbf/in² = 6.894 757 kPa	1 kPa = 0.15 lbf/in²
bearing pressures	kilogram-force/centimeter²	kilopascal (kPa)	1 kgf/cm² = 98.0665 kPa	1 kPa = 0.01 kgf/cm²
Barometric measurements and pressures of small magnitude, differential pressures, ventilation pressures	inch of mercury (60°F)	kilopascal (kPa)	1 inHg = 3.376 85 kPa	1 kPa = 0.3 inHg
	inch of water (60°F)	kilopascal (kPa)	1 inH_20 = 0.248 84 kPa	1 kPa = 4.0 inH_20
3-11.2 Stress				
Stresses in materials	pound-force/inch²	megapascal (MPa)	1 lbf/in² = 6.894 757 x 10^{-3} MPa	1 MPa = 145.4 lbf/in²
	ton-force/inch²	megapascal (MPa)	1 tonf/in² = 15.444 26 MPa	1 MPa = 0.065 tonf/in²
	kip/inch²	megapascal (MPa)	1 kip/in² = 6.894 757 MPa	1 MPa = 0.145 kip/in²

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CONVERSION TABLE—Continued

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present (2)
3-19.1	Dynamic Viscosity				
	Viscosity of liquids and gases	centipoise (cP)	millipascal second (mPa•s)	1.0*	1.0*
		pound force second per square foot	pascal second (Pa•s)	1 lbf•s/ft² = 47.880 26 Pa•s	1 Pa•s = 0.02 1 lbf•s/ft²
3-20.1	Kinematic Viscosity				
	Viscosity of liquids	centistokes (cSt)	square millimeter per second (mm²/s)	1.0*	1.0*
		square foot per second	square meter per second (m²/s)	1 ft²/s = 0.092 903 m²/s	1 m²/s = 10.76 ft²/s
3-22.1	Energy, Work				
	Thermodynamics	Btu	kilojoule (kJ)	1 Btu = 1.055 056 kJ	1 kJ = 0.95 Btu
		foot pound force	joule (J)	1 ft•lbf = 1.355 818 J	1 J = 0.738 ft•lbf
	Battery rating	ampere-hour	ampere hour (A•h)	1.0*	1.0*
3-23.1	Power				
	Mechanical, electrical or hydraulic power of normal magnitude	horsepower (SHP, BHP, etc.)	kilowatt (kW)	1 hp = 0.745 700 kW	1 kW = 1.34 hp
	Very small powers	foot found force/second	watt (W)	1 ft•lbf/s = 1.355 818 W	1 W = 0.738 ft•lbf/s
4-1.1	Absolute Temperature				
	Thermodynamic calculations	degree Rankine	kelvin (K)	1 °R = 9/5 K*	K = 5/9 °R*

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4-2.1 Temperature				
All measured and specified temperature	degree Fahrenheit	degree Celsius (°C)	$t°_F = 9/5t°_C + 32$	$t°_C = (5/9)(t°_F - 32)$
4-2.1a Temperature Interval				
All temperature intervals	degree Fahrenheit	degree Celsius (°C)	1 °F = 5/9 °C*	1° C = 9/5 °F
			1 °F = 5/9 K*	1 K = 9/5 °F
4-3.1 Linear Expansion Coefficient				
All linear expansion coefficients	inch/inch deg F	meter per meter K (K^{-1})	1 $°F^{-1}$ = 9/5 K-1*	1 k^{-1} = 5/9 $°F^{-1}$
4-4.1 Heat, Quantity of Heat				
Thermodynamics, refrigeration	Btu	kilojoule (kJ)	1 Btu = 1.055 056 kJ	1 kJ = 0.948 Btu
4-5.1 Heat Flow Rate				
Boiler and heat exchanger design	Btu/second	kilowatt (kW)	1 Btu/s = 1.055 056 kW	1 kW = 0.948 Btu/s
Thermodynamics, Refrigeration and air conditioning	Btu/hour	watt (W)	1 Btu/h = 0.293 071 W	1 W = 3.4 Btu/h
	ton of refrigeration	kilowatt (kW)	1 ton (refrig) = 3.1568 kW	1 kW = 0.317 ton (refrig)
4-5.1a Density of Heat Flow Rate				
Boiler and heat exchanger design	Btu/foot² hour	watt per square meter (W/m^2)	1 Btu/(ft^2·h) = 3.154 591 W/m^2	1 W/m^2 = 0.32 Btu/(ft^2·h)
4-5.1b Volumetric Heat Release Rate				
Boiler design	Btu /(ft^3·h)	watt per cubic meter (W/m^3)	1 Btu/(ft^3·h) = 10.349 W/m^3	1 W/m^3 = 0.097 Btu/(ft^3·h)

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CONVERSION TABLE—Continued

ISO No.	Quantity/areas of use	Present units	Recommended SI units(1)	Conversion factors: Present to recommended	Conversion factors: Recommended to present (2)
4-5.1c	Fuel Consumption Rate Specific Fuel Consumption Heat engine design, thermodynamics	pound/horsepower hour	kilogram per kilowatthour kg/(kW•h)	1 lb/hph = 0.608 278 kg/(kW•h) 1 lb/hph = 1.689 659 x 10^{-7} kg/J	1 kg/(kW•h) = 1.64 lb/hph 1 kg/J = 59.2 x 10^{5} lb/hph
4-7.1	Thermal Conductivity Heat exchanger design, insulation	Btu inch/foot² hour Degree F	watt per square meter °C W/(m²•°C)	1 Btu•in/(ft²•h•°F) = 0.144 228 W/m²•°C)	1 W(m²•°C) = 6.93 Btu•in/(ft²•h•°F)
4-8.1	Coefficient of Heat Transfer Heat exchanger design, insulation	Btu/hour-foot² deg F	watt per square meter °C W/(m²•°C)	1 Btu/(h•ft²•°F) = 5.678 263 W/m²•°C)	1 W(m²•°C) = 0.176 Btu/(h•ft²•°F)
4-11.1	Specific Heat Capacity Thermodynamics, applied heat, refrigeration	Btu/pound °F	kilojoule per kilogram degree C kJ/(kg•°C).	1 Btu/(lb•°F) = 4.1868 kJ/(kg•°C)*	1 kJ/(kg•°C) = 0.24 Btu/(lb•°F)
4-14.1	Specific Entropy Thermodynamics	Btu/pound degree R.	kilojoule per kilogram degree K kJ/(kg•K).	1 Btu/(lb•R) = 4.1868 kJ/(kg•K)*	1 kJ/(kg•K) = 0.24 Btu/(lb•°R)
4-16.1	Specific Energy, Calorific Value, Specific Latent Heat thermodynamics, refrigeration	Btu/pound	kilojoule per kilogram (kJ/kg)	1 Btu/lb = 2.326 kJ/kg*	1 kJ/kg = 0.43 Btu/lb

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5-1.1 Current	ampere	ampere (A)	1.0*		1.0*
5-2.1 Quantity of Electricity, electrical charge	coulomb	coulomb (C)	1.0*		1.0*
5-6.2 Electromotive Force Potential Difference	volt	volt (V)	1.0*		1.0*
5-11.1 Capacitance	farad	farad (F)	1.0*		1.0*
5-21.1 Magnetic Field Intensity	oersted	ampere per meter (A/m)	1 oersted = 79.58 A/m		1 A/m = 0.0126 oersted
5-24.1 Magnetic Flux Density	gauss	tesla (T)	1 gauss = 10^{-4} T*		1 T = 1000 gauss*
5-25.1 Magnetic Flux	maxwell	weber (Wb)	1 maxwell = 10^{-8} Wb*		1 Wb = 10^{7} maxwell*
5-26.1 Magnetomotive Force	gilbert	ampere (A)	1 gilbert = 0.7958 A		1 A = 1.257 gilbert
5-27.1 Inductance	henry	henry (H)	1.0*		1.0*
5-29.1 Permeability	gauss/oersted	henry per meter (H/m)	1 gauss/oersted = 1.257×10^{-6} H/m		1 H/m = 7.96×10^{5} gauss/oersted
5-31.1 Resistance	ohm	ohm (Ω)	1.0*		1.0*
5-42.1 Conductance	mho	siemens (S)	1 mho = 1 S*		1.0*
5-43.1 Resistivity	ohm-centimeter (Ω•cm)	ohm-meter (Ω•m)	1 Ω•cm = 0.01 Ω•m*		1 Ω•m = 100 Ω•cm*
	ohm-circular mil/foot	ohm millimeter² per meter	1 Ω•cmil/ft = 0.001 662 Ω•mm²/m		1 Ω•mm²/m = 601.68 Ω•cmil/ft
5-44.1 Conductivity	mho/centimeter	siemens per meter (S/m)	1 mho/cm = 100 S/m*		1 S/m = 0.01 mho/cm*
5-52.1 Electric Power	watt	watt (W)	1.0*		1.0*
6-19.1 Luminous Intensity	candlepower	candela (cd)	1 candlepower = 1 cd*		1.0*
6-20.1 Luminous Flux	lumen	lumen (lm)	1 lumen = 1 cd•sr*		1.0*
6-22.1 Luminance	lambert	candela per meter² (cd/m²)	1 lambert = 3183 cd/m²		1 cd/m² = 3.14×10^{-4} lambert
6-24.1 Illuminance	footcandle	lux (lx)	1 footcandle = 10.76 lx		1 lx = 0.093 footcandle
6-50.1 Activity (of a radionuclide)	curie	becquerel (Bq)	1 curie = 3.7×10^{10} Bq*		1 Bq = 2.7×10^{-11} curie

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* Note: Indicates exact conversion value.

(1)This table lists the preferred metric units (SI units and units accepted for use with SI) recommended for use throughout the Maritime Industry.

(2)Rounded. For greater precision, use the reciprocal of the "Present to Recommended" factor.

(3)Minutes and seconds may still be used for Navigation purposes.

(4)For values of nautical mile and knot, see table VI page 5.

(5)The volume measurement for those not using the metric system is generally 40 cubic feet, however, 50 cubic feet is used occasionally. It is recommended that the cubic meter (35.314 cubic feet) be used as the replacement unit. The conversion factor or one non metric measurement ton (40 cu ft) equals 1.132 67 cubic meters. One cubic meter of 35.314 cubic feet equals 0.883 non metric measurement tons. The conversion of tariffs filed with the Federal Maritime Commission must be accomplished in accordance with their filing requirements.

(6)All tonnage measurements are long tons (2240 lb.). The short ton (2000 lb.) equals 0.907 185 t.

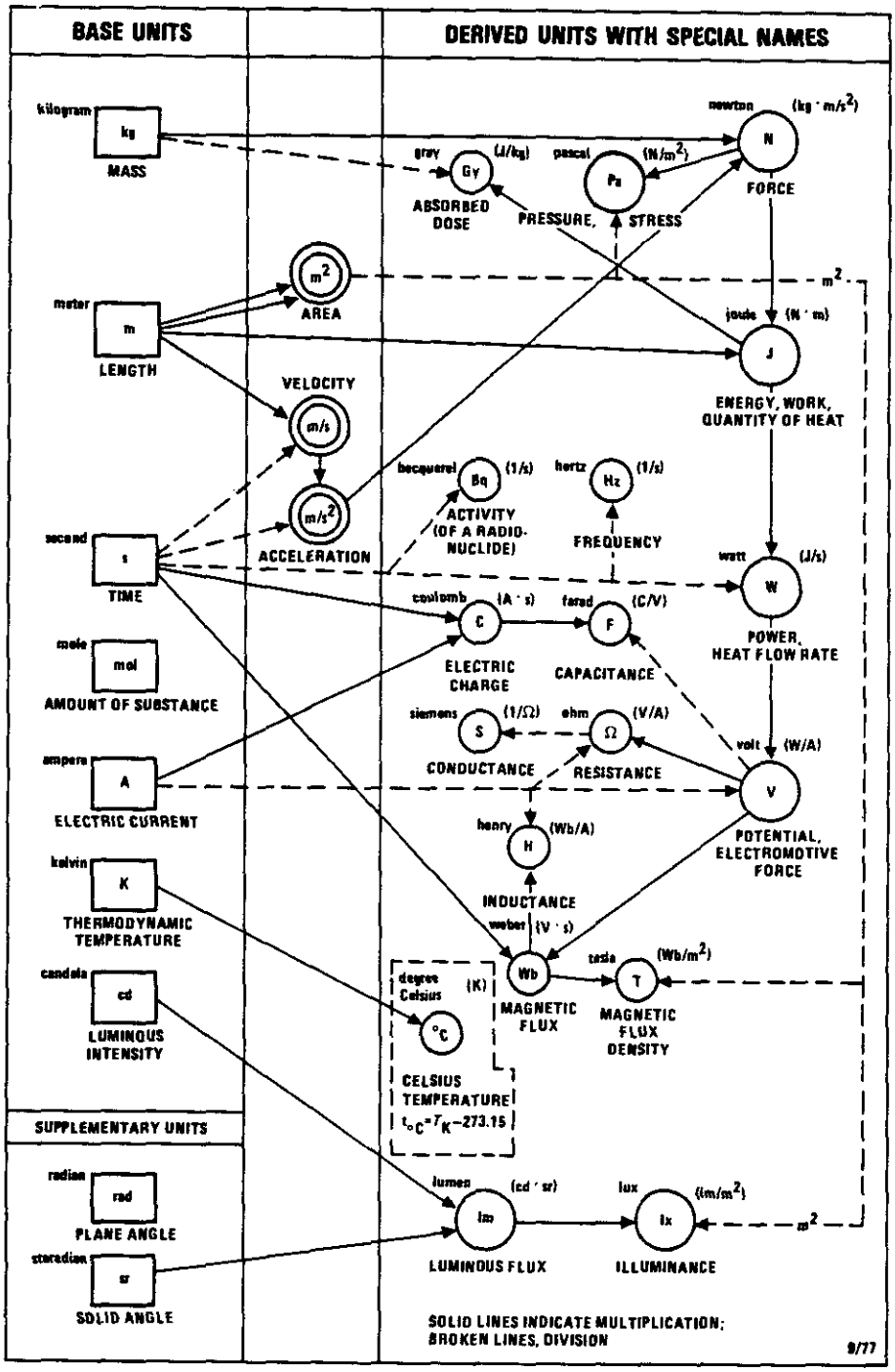
(7)An apparent anomaly exists in the case of the joule for work (J = N·m) and the use of N·m for torque or bending moment. These are, however, entirely different units. In the former, the unit of work results from unit force moving through unit distance. In the latter unit force acts at right angles to the lever arm of unit length. This would be readily seen if vectors were incorporated in the unit symbols. For these reasons, it is important to express work or energy in joules and moment of force or torque in newton meters, not joules. The use of newton meter per radian as the unit of torque in rotational dynamics will result in dimensional integrity.

(8)It has been internationally recommended that pressure units themselves should not be modified to indicate whether pressure is "absolute" (that is, above zero) or "gage" (that is, above atmospheric pressure). If, therefore, the context leaves any doubt as to which is meant, the word "pressure" must be qualified appropriately.

". . . at a gage pressure of 13 kPa" or
". . . at an absolute pressure of 13 kPa" or
". . . reached an absolute pressure of 13 kPa".
etc.

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RELATIONSHIPS OF SI UNITS WITH NAMES



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ERRATA TO THE APPENDIX

Page 5, Table III

1. In heading, for last column, correct "Base" to read "base".
2. For "Celsius temperature interval", in "Symbol" column, correct "° C" to read "°C"; in last column, correct "1 °C" to read "1°C".

Page 7, Table VI

1. For "nautical mile", after "1.852 km", insert "(exact)".
2. For "knot", correct "hr." to read "h".

Page 8, in 4.1.1

1. Correct second sentence to read "When using systems with limited character sets, as in Telex transmission or computer printout, the standard symbols cannot be used." (Note: Complete line had been omitted.)

Page 10, in 5.4

1. In last line on page, insert "cubic" before "meters".

Page 12

1. In fourth paragraph, in fifth line, correct spelling of "ratio".
2. In last paragraph, third sentence, delete comma after "C_v".

Page 17, for 3-1.1c

1. Insert "Unit" before "Displacement".

Page 19, in 1-1.1

1. In "Recommended SI units" column, for second entry, delete asterisk after "degree (°C)".
2. In last column, insert asterisks after "minute" and "second" conversions.
3. In last column, delete period after "min".
4. In last two columns, correct "sec" and "sec." to read "s".

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Page 21, in 1-5.1

1. For "measurement ton", in last column, correct "35.31" to read "0.3531".

Page 21, in 1-5.1b

1. For "Capacity of most pumps" second line, in last column, correct "15 873.0" to read 15 850".
2. For "Barrel per nautical mile", in last two columns, correct "bbl/mi" to read "bbl/nmi".

Page 22, in 1-6.1

1. In "Present units" column, in third line, correct "(m)" to read "(min)".

Page 22, in 2-3.1

1. In "Quantity" column, insert "Electrical and" before "Electronics".

Page 23, in 3-1.1c

1. In first line, insert "Unit" before "Displacement".
2. In second line, in "Present units" column, correct "tons per inch" to read "ton per inch".

Page 24, in 3-2.1a

1. In first "Conversion factors" column, correct "0.17 118" to read "0.017 118".

Page 25, in 3-10.1

1. In first column, delete bottom line reading "Torque(7)"; insert footnote 7 indicator in first line after "(Torque)".

Page 26, in 3-22.1

1. Move entire "Battery rating" line to 5-2.1.

Page 26, in 4-1.1

1. In first "Conversion factors" column, correct "1 °R" to read "1°R".
2. In last column, insert "1" before "K".

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Page 27, in 4-2.1

1. In last two columns, insert asterisk at end of equations.

Page 27, in 4-2.1a

1. In first "Conversion factors" column, correct "1 °F" to read "1°F" (twice).
2. In first line, in last column, correct "1 °C" to read "1°C".
3. In last column, insert asterisk at end of equation (both lines).

Page 27, in 4-3.1

1. In first "Conversion factors" column, correct equation to read "$1°F^{-1} = 9/5\ K^{-1}$*".
2. In last column, correct "1 k^{-1}" to read "1 K^{-1}"; insert asterisk at end.

Page 28, in 4-5.1c

1. In "Recommended SI units" column, enclose symbol in parentheses "(kg/(kW•h))".
2. In "Recommended SI units" column, in line corresponding to second set of conversion equations, insert "kilogram per joule (kg/J)".

Page 28, in 4-7.1

1. In "Recommended SI units" column, correct unit to read "watt per meter degree C (W/(m•°C))".
2. In both conversion equations, correct "m^2" to read "m".

Page 28, in 4-8.1

1. In "Recommended SI units" column, correct unit to read "watt per square meter degree C (W/(m^2•°C))".

Page 28, in 4-11.1

1. In "Recommended SI units" column, correct unit to read "kilojoule per kilogram degree C (kJ/(kg•°C)).

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Page 28, in 4-14.1

1. In "Recommended SI units" column, correct unit to read "kilojoule per kilogram kelvin (kJ/(kg·K))".
2. In first "Conversion factors" column, correct "R" to read "°R".

Page 29, in 5-2.1

1. As second item, insert entire line "Battery rating ....", moved from 3-22.1.

Page 29, in 5-6.2

1. Insert comma between "Force" and Potential".

Page 30, in Note 5

1. In third sentence, correct fourth word "or" to read "for".
2. In third and fourth sentence, correct "non metric" to read "non-metric".

(Note: Throughout Table VII, because of crowded conditions, the horizontal dotted line occasionally appears as a single dot; care should be taken that it is not misconstrued as an on-the-line decimal marker.)

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